

Medical Students' Perceptions of Virtual Reality (VR) in Anatomy Education: A Pilot Study in Rwanda

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Abstract

Background:

Anatomy teaching in Rwanda largely depends on lectures and cadaveric dissections, with limited use of digital tools. To meet Rwanda's 2030 healthcare workforce goals, innovative teaching methods are needed. Virtual Reality (VR) offers an immersive, scalable platform that addresses shortages in cadavers, classrooms, and staff. MediXR, a Rwandan startup, developed a VR anatomy simulator to enhance learning. This pilot evaluated medical students' perceptions of VR-based anatomy education.

Methodology:

A one-day workshop at IRCAD Africa, Kigali, involved 25 participants from four medical schools (University of Rwanda [UR], Rwanda Military Academy [RMA], Adventist School of Medicine [ASOME], and University of Medical Sciences and Technology [UMST]). Participants explored an upper-limb VR anatomy module developed by MediXR and validated by UR. Post-session feedback was collected electronically. While 25 responses were recorded, only 22 entries were retained for matched pre- and post-workshop analysis due to missing or inconsistent data. Descriptive and inferential analyses were conducted using Microsoft Excel and Python

(replicating SPSS functions). Satisfaction, perceived relevance, and learning gain were assessed using Likert scales.

Results:

Of the 22 analyzed participants, 14 were male (56%) and 11 female (44%). Institutions represented were UR (24%), RMA (20%), ASOME (16%), and UMST (40%). Most participants were MEDSAR members (16/22, 73%). Thirteen participants reported no prior VR experience. Overall satisfaction was high (Median = 5, IQR = 0), workshop content relevance was rated high (Median = 4, IQR = 0.75), and learning gain in upper-limb anatomy was notable (Median = 3, IQR = 1). Statistical analysis revealed no significant differences in satisfaction between participants with or without prior VR experience (Mann-Whitney U = 62.5, $p = 0.727$, $r = -0.075$). Differences in learning gain across institutions were also not statistically significant (Kruskal-Wallis H = 6.62, $p = 0.158$, $\eta^2 = 0.201$). Correlation analysis indicated moderate positive associations between interest in surgery and post-workshop relevance ($r = 0.39$) and learning gain ($r = 0.28$), and a strong correlation between satisfaction and learning gain ($r = 0.49$).

Conclusion:

The pilot demonstrated strong student enthusiasm and perceived learning gains from VR anatomy instruction. Satisfaction and engagement were high, suggesting VR is a feasible and effective supplement to traditional teaching in low-resource settings. Limitations include small sample size, reliance on self-reported data, and partial dataset matching. Future larger studies should objectively measure knowledge retention and explore long-term integration of VR into Rwandan medical education.

Keywords: Virtual reality, anatomy education, Rwanda, simulation, medical training

Introduction

Anatomy forms the cornerstone of medical education and clinical competence in Rwanda [1]. Many students perceive anatomy as cognitively challenging, which may contribute to gaps in clinical application [2–4]. While cadaveric dissection remains the gold standard [5], Rwanda faces significant constraints due to limited cadaver donations [1].

The Government of Rwanda's 4×4 Health Sector Reform Plan aims to quadruple the healthcare workforce by 2030 [6]. However, infrastructure expansion has not matched increased admissions. Some institutions use 3D learning technologies (e.g., Anatomage Table) to bridge gaps [1], though these are costly and not easily scalable.

MediXR, a Rwandan startup, developed a VR anatomy solution as an affordable, immersive alternative. VR enables users to interact with realistic 3D environments and has been shown globally to enhance engagement, understanding, and confidence in medical learners [7–10].

Despite evidence elsewhere, VR's effectiveness in Rwandan anatomy education has not been formally evaluated. This pilot study aimed to assess medical students' reception, perceived learning, and educational impact of VR-based anatomy instruction in Rwanda.

Methodology

Study Site

The study was conducted at the Institut de Recherche contre les Cancers de l'Appareil Digestif (IRCAD) in Kigali, Rwanda. IRCAD is a regional surgical training and research center equipped with advanced simulation and imaging technologies, providing an ideal environment for immersive learning experiences.

Study Design

This pilot cohort study adopted a one-day interactive workshop format themed "From Practice to Precision: Immersive Learning for Future Doctors." The event was organized by MediXR, a Rwandan health technology startup, in collaboration with MEDSAR (Medical Students' Association of Rwanda), SCOME (Standing Committee on Medical Education), and IRCAD Africa on July 18, 2025.

The primary objective was to evaluate the feasibility, reception, and perceived educational impact of a locally developed Virtual Reality (VR) anatomy tool among undergraduate medical students in Rwanda.

Study Population and Sampling

Of the 127 medical students who registered online, 25 participants were purposively selected to ensure gender balance and representation across Rwanda's four medical schools (UR, ASOME, UMST, and RMA). As this was a pilot study focused on evaluating usability and initial user perceptions, formal sample size calculations were not required.

This sample size aligns with standard recommendations for pilot studies of educational interventions [Julious SA, 2005].

Inclusion Criteria:

- Undergraduate medical students who completed the Limbs and Back Anatomy module
- Students who confirmed availability and provided informed consent.

Exclusion Criteria:

- Students who had not completed the upper limb anatomy module.

- Those who did not consent to participate.

Workshop Structure

The workshop commenced with opening remarks by guest speakers from the Ministry of ICT and Innovation, Imisi 3D, and the Dean of the School of Medicine and Pharmacy, University of Rwanda. A didactic lecture on upper limb anatomy (brachial plexus) was delivered virtually by Prof. Julien Gashegu, Chair of the Department of Human Anatomy, UR. This was followed by hands-on VR sessions and an interactive quiz competition.

Participants were divided into five groups (n=5) to facilitate peer-assisted learning. Each participant used the MediXR VR tool to interact with 3D models of the bones, muscles, and neurovascular structures of the upper limb.

Following the session, participants completed a post-workshop feedback form to assess perceived learning impact, usability, and satisfaction.

Data Collection

Pre- and post-workshop data were collected using Google Forms:

- **Pre-workshop survey:** Captured demographic information, previous exposure to VR, motivation for participation, and learning expectations.
- **Post-workshop survey:** Captured self-reported satisfaction, perceived improvement in understanding, and qualitative feedback regarding the most engaging aspects of the workshop.

Ethical Considerations

Ethical approval was obtained from King Faisal Hospital IRB (Ref: KFH/2024/181/IRB). Informed consent was obtained electronically. Informed consent was obtained electronically before participation. Confidentiality and anonymity were ensured by excluding identifiable information from all datasets and reports.

Data Analysis

Descriptive statistics summarized demographic and outcome variables. Likert-scale data were reported as medians and interquartile ranges (IQR). Non-parametric tests (Mann-Whitney U, Kruskal-Wallis H) assessed group differences. Correlations between pre-workshop interests and post-workshop outcomes were examined using Pearson correlation.

Results

Sociodemographic Characteristics

Of the 25 participants, 14 were male (56%) and 11 female (44%). Institutional representation included UR (24%), RMA (20%), ASOME (16%), and UMST (40%). Majority were MEDSAR members (16/22, 73%). Academic years of participants were in preclinical class two (37.8%) and preclinical class one (36.2%). Others were in year three (9.4%), year four (10.2%), year five (3.1%), and unspecified (3.1%) (Figure 1).

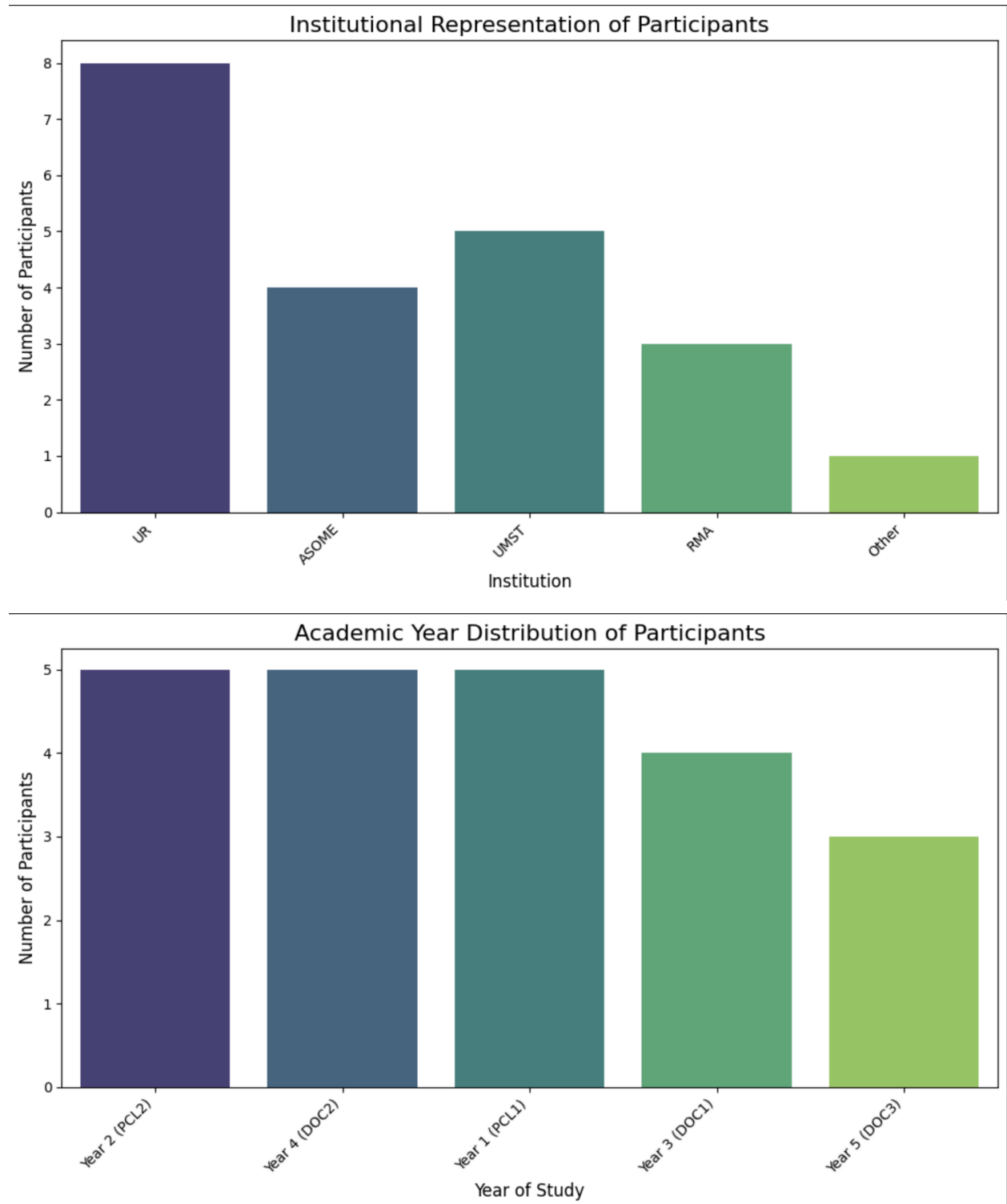


Figure 1: Bar chart of participant distribution by institution and academic year.

Prior VR Experience and Interests

13 of 52% of participants reported no prior experience with VR tools before the workshop. Despite this limited exposure, nearly all participants expressed a strong interest in adopting VR for anatomy and medical education within their respective institutions.

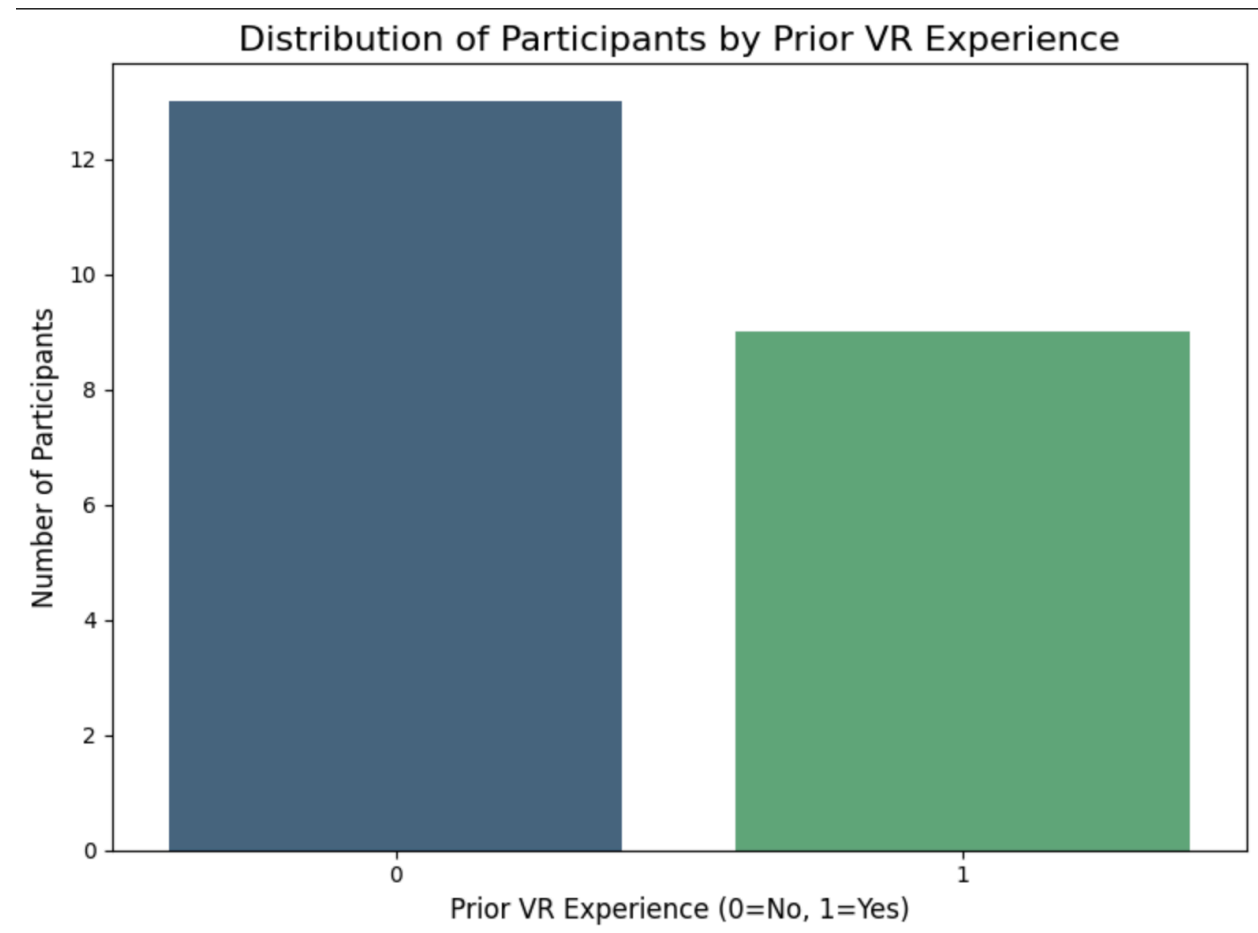


Figure 2: Bar chart showing prior VR experience of participants.

Post-Workshop Satisfaction and Learning

All participants reported high satisfaction (Median = 5, IQR = 0). Relevance of workshop content was rated high (Median = 4, IQR = 0.75), and learning gain in upper-limb anatomy was notable (Median = 3, IQR = 1).

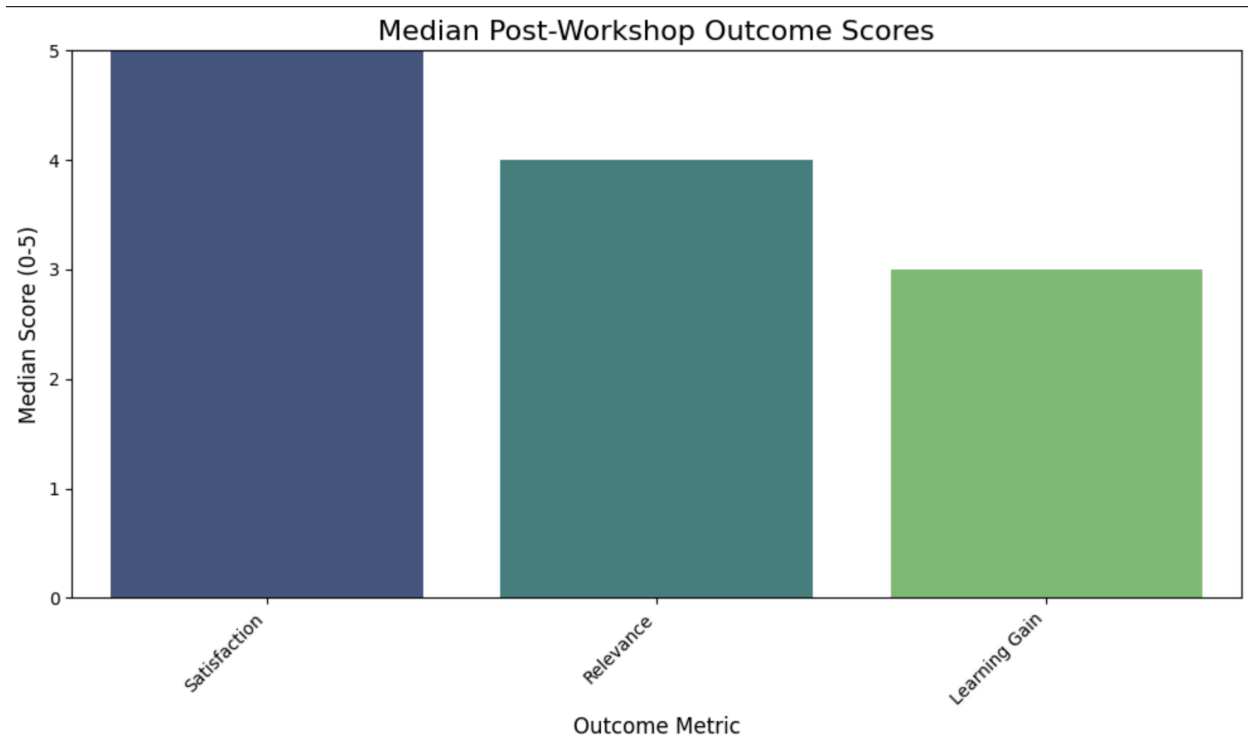


Figure 3: Bar chart of post-workshop outcomes (satisfaction, relevance, learning gain).

Inferential Statistics:

- Mann-Whitney U test: Satisfaction vs. prior VR experience, $U = 62.5$, $p = 0.727$, $r = -0.075$
- Kruskal-Wallis H test: Learning gain vs. institution, $H = 6.62$, $p = 0.158$, $\eta^2 = 0.201$
- Correlations: Interest in surgery vs. relevance ($r = 0.39$), vs. learning gain ($r = 0.28$); satisfaction vs. learning gain ($r = 0.49$)

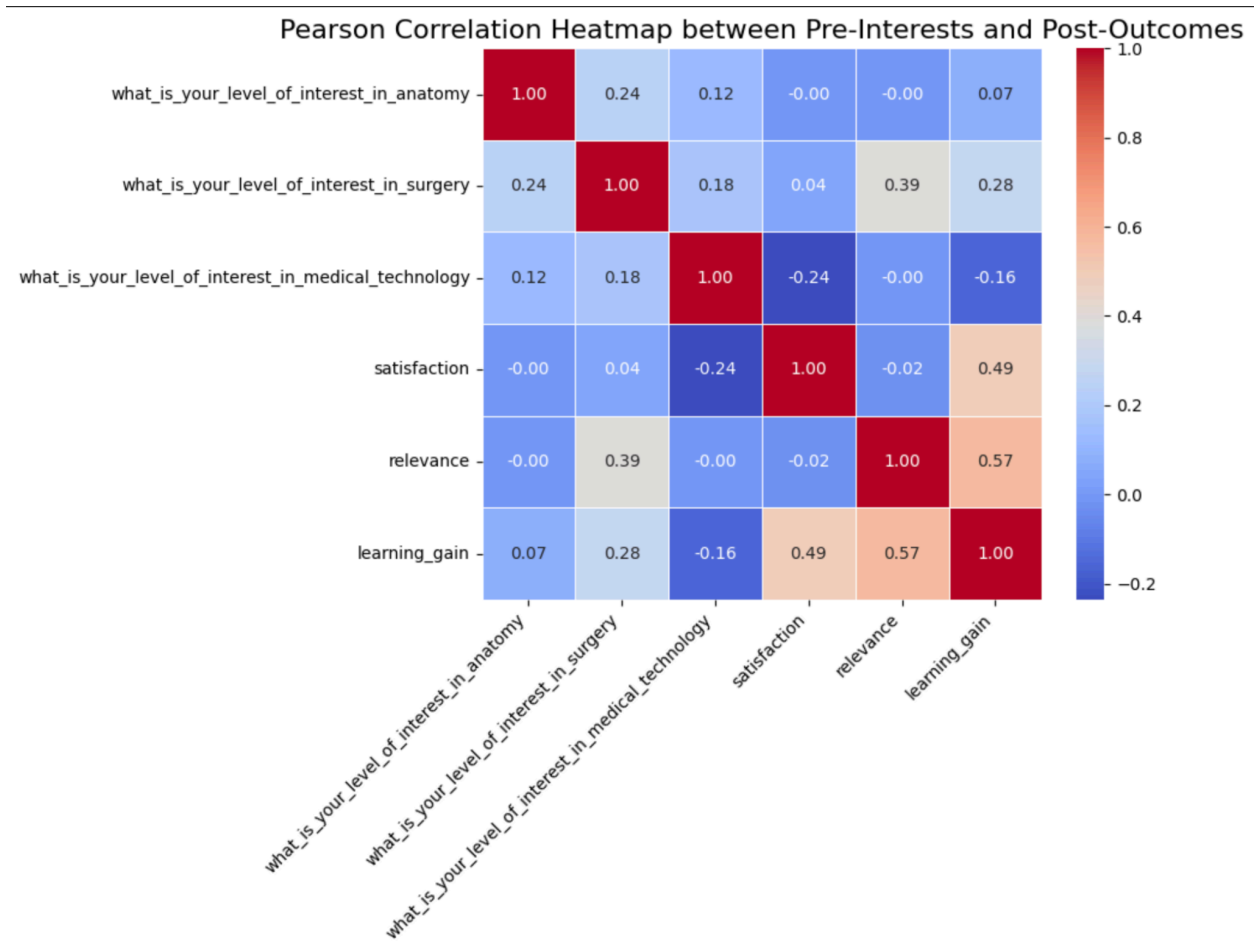


Figure 4: *Heatmap of correlations between pre-workshop interest and post-workshop outcomes.*

Qualitative Feedback

Participants described VR as “transformative,” “interactive,” and “eye-opening.”

The most frequently suggested improvements included: Extending the duration of VR sessions; Incorporating additional clinical modules; Organizing follow-up workshops and faculty training for sustainable implementation

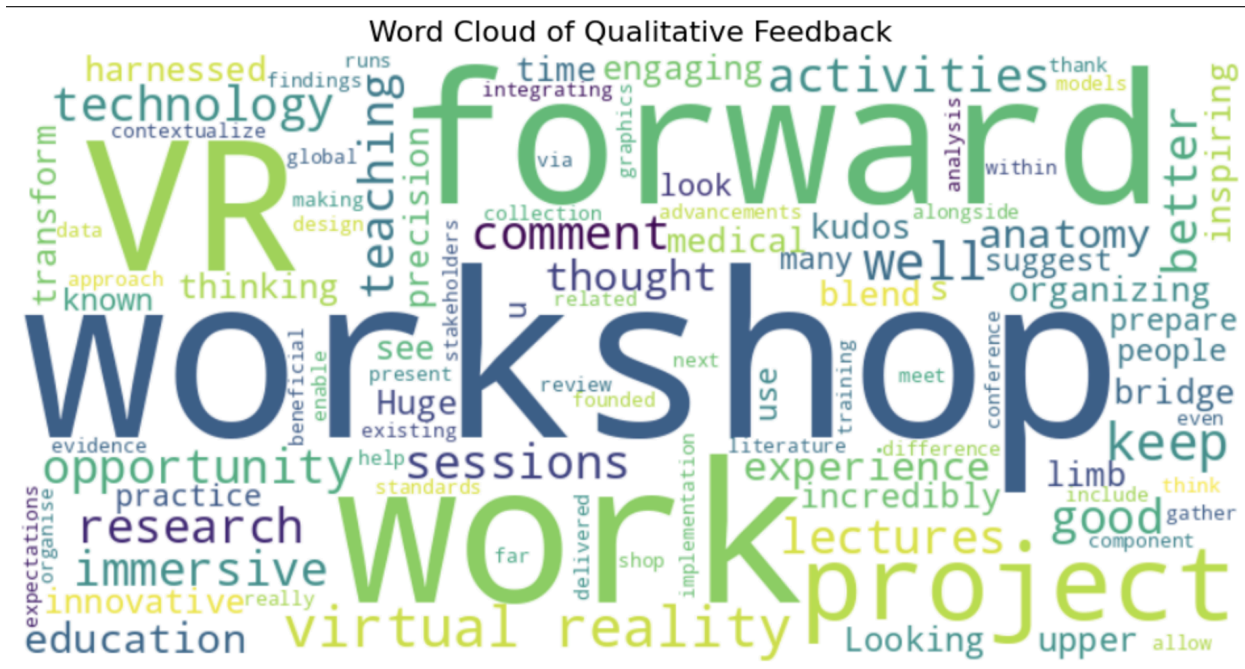


Figure 5: *Word cloud or thematic summary of qualitative feedback.*

Discussion

The study demonstrates strong acceptance of VR-based anatomy instruction in Rwanda, with 100% reporting improved understanding of the upper limb and high satisfaction (92.3%). Moderate correlations suggest pre-existing interest in surgery enhances engagement and learning outcomes, while a strong correlation between satisfaction and learning gain highlights the impact of a positive workshop experience.

Findings align with global literature showing immersive tools improve spatial understanding and retention [Jannin, 2019; Zhang et al., 2018]. MediXR's VR simulator enables interactive exploration of 3D anatomical structures, bridging theory and clinical practice.

Advocacy for curriculum integration (81.3%) reflects growing readiness for digital transformation. VR offers scalable solutions to barriers such as limited cadavers, overcrowded labs, and instructor shortages. A blended model combining VR and traditional methods may optimize learning efficiency.

Limitations

- Small sample size ($n = 22$ analyzed) limits generalizability and reduces statistical power.
- Reliance on self-reported measures introduces potential bias.
- Ordinal scales limit granularity compared to continuous measures.

- Effect size approximations (Mann-Whitney) may be less precise due to small N.
- Deduplication and merging may have excluded some participant data, affecting representativeness.
- VR cannot replicate tactile feedback, highlighting the need for hybrid training models.

Conclusion

This pilot study provides preliminary evidence that immersive VR-based instruction is a feasible and effective supplementary tool for anatomy education in Rwanda. The MediXR platform enhanced engagement, satisfaction, and perceived understanding among medical students from diverse institutions. Integrating such technologies into anatomy curricula could transform medical education by improving accessibility, scalability, and learner-centered experiences in resource-limited settings. However, VR should complement, not replace, cadaveric dissection, ensuring that the development of cognitive, spatial, and manual competencies remains balanced. Future large-scale studies are recommended to validate these findings and explore the long-term educational impact of VR-based anatomy learning across sub-Saharan Africa.

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Conflict of Interest

None declared

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References

1. Archibong VB, et al. Anatomage integration in medical education. *J Med Educ.* 2025;13:1–10.
2. Almizani MS, et al. Clinicians' and students' perceptions and attitudes regarding the anatomical knowledge of medical students. *Adv Med Educ Pract.* 2022;13:1251–1259.
3. Feigl G, Sammer A. The influence of dissection on clinical anatomical knowledge for surgical needs. *Surg Radiol Anat.* 2022;44:9–14.
4. Cheung CC, et al. Why is anatomy difficult to learn? The implications for undergraduate medical curricula. *Anat Sci Educ.* 2021;14(6):752–763.
5. Allen MA, Kirkpatrick N, Agosto ER. Anatomage Table 6. *J Electron Resour Med Libr.* 2019;16:59–66.
6. Ministry of Health Rwanda. 4×4 Health Workforce Reform. Kigali: MOH; 2024.
7. Zhang M, et al. Recent developments in game-based virtual reality educational laboratories using the Microsoft Kinect. *Int J Emerg Technol Learn.* 2018;13:138–159.
8. Gadelha R. Revolutionizing education: The promise of virtual reality. *Child Educ.* 2018;94:40–43.
9. World Medical Association. Declaration of Helsinki. *JAMA.* 2024;331:2179–2184.
10. ALU students pioneer VR medical training. *TopAfricaNews.* June 2024.
11. MediXR internal dataset: Medical virtual reality participants list, workshop application and feedback responses; 2025.
12. Jiang H, et al. Virtual reality in medical students' education: Scoping review. *JMIR Med Educ.* 2022;8:e36441.
13. Sibomana O. Could virtual reality be a solution in surgical trainings in resource-restricted settings? *Surg Open Sci.* 2024;24:14–16.
14. Neher AN, et al. Virtual reality for assessment in undergraduate nursing and medical education – a systematic review. *BMC Med Educ.* 2025;25:292.
15. Julious SA. Sample size of 12 per group rule of thumb for a pilot study. *Pharm Stat.* 2005;4(4):287–291.